

Machine Learning

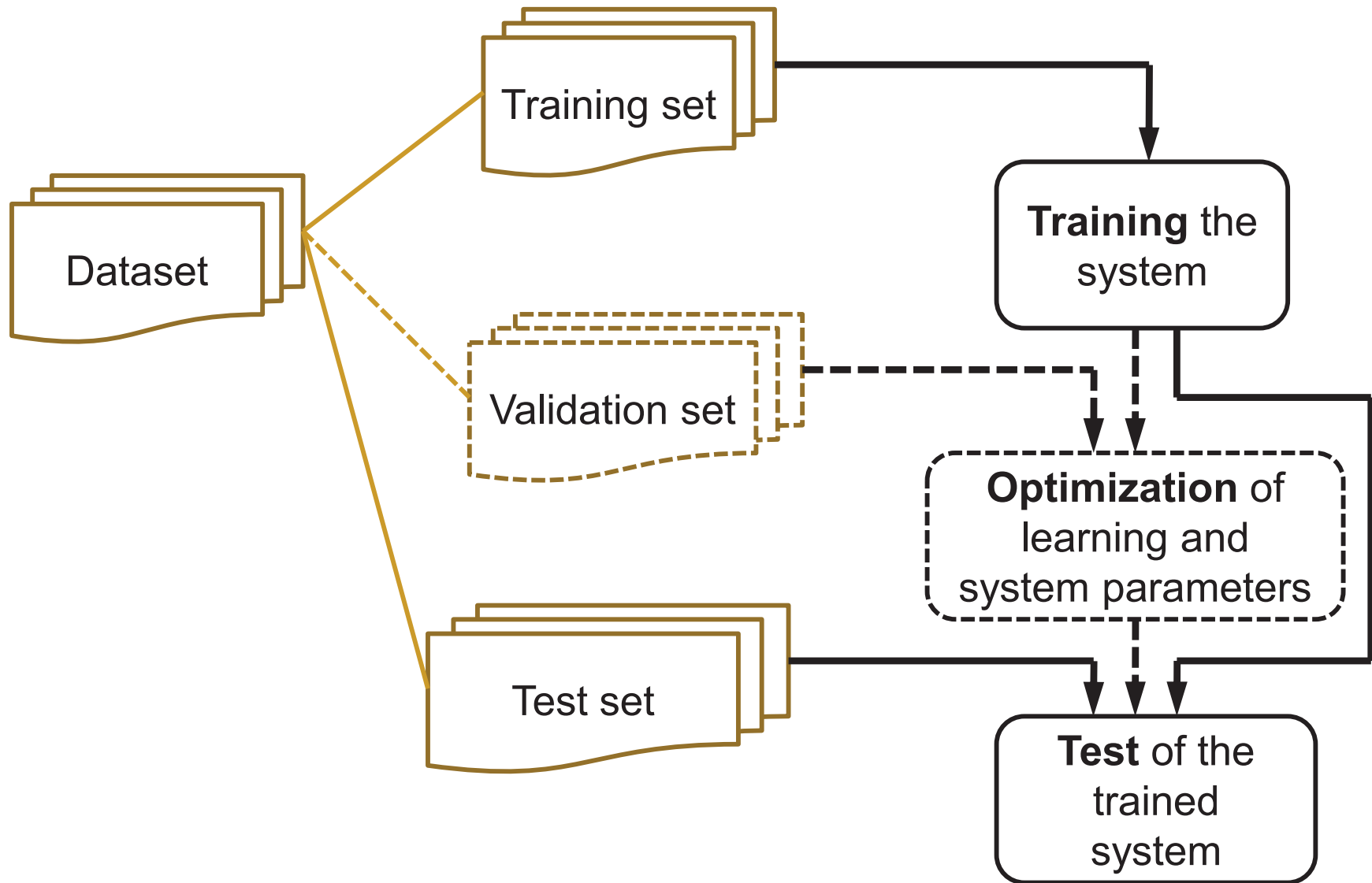
(IT3190E)

Quang Nhat NGUYEN

quang.nguyennhat@hust.edu.vn

Hanoi University of Science and Technology
School of Information and Communication Technology
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Machine learning process



Main elements of ML problem (1)

■ Training (learning) examples

- The training feedback is included in training examples or indirectly provided (e.g., from the working environment)
- They are supervised or unsupervised training examples
- The training examples should be compatible with (i.e., representative for) the future test examples

■ The target function to be learned

- $F: X \rightarrow \{0,1\}$
- $F: X \rightarrow$ A set of class labels
- $F: X \rightarrow \mathbb{R}^+$ (i.e., a domain of positive real values)
- ...

Main elements of ML problem (2)

- Representation of the target function to be learned
 - A polynomial function
 - A set of rules
 - A decision tree
 - An artificial neural network
 - ...
- ML algorithm that can learn *approximately* the target function
 - Regression-based
 - Rule induction
 - Decision tree learning (e.g., ID3 or C4.5)
 - Back-propagation
 - ...

Challenges in ML (1)

■ Learning algorithm

- Which learning algorithms can learn approximately a given target function?
- Under which conditions, a selected learning algorithm converges (approximately) the target function?
- For a specific application problem and a specific example (object) representation, which learning algorithm performs best?

Challenges in ML (2)

■ Training examples

- How many training examples are enough for the training?
- How does the size of the training set (i.e., the number of training examples) affect the accuracy of the learned target function?
- How do error (noise) and/or missing-value examples affect the accuracy?

Challenges in ML(3)

■ Learning process

- What is the best ways of use order of training examples?
- How does the order of using training examples vary the complexity of the ML problem?
- How does the application problem-specific knowledge (apart from the training examples) contribute to the machine learning process?

Challenges in ML (4)

- Learning capability
 - Which target function the system should learn?
 - Representation of the target function: Representation capability (e.g., linear / non-linear function) vs. Complexity of the learning algorithm and learning process
 - The theoretical limits for the learning capability of learning algorithms?
 - The system's capability of generalization from the training examples?
 - **Under-fitting** problem
 - **Over-fitting** problem
 - The system's capability of self-adapting its internal architectural representation?
 - To improve the system's capability of representation and learning of the target function

Challenges in ML (5)

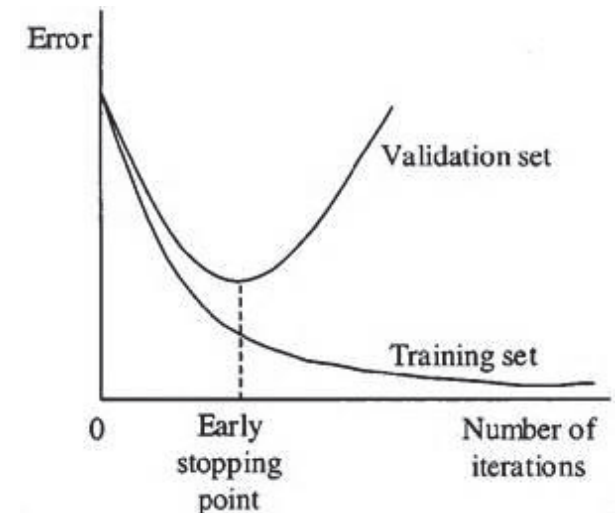
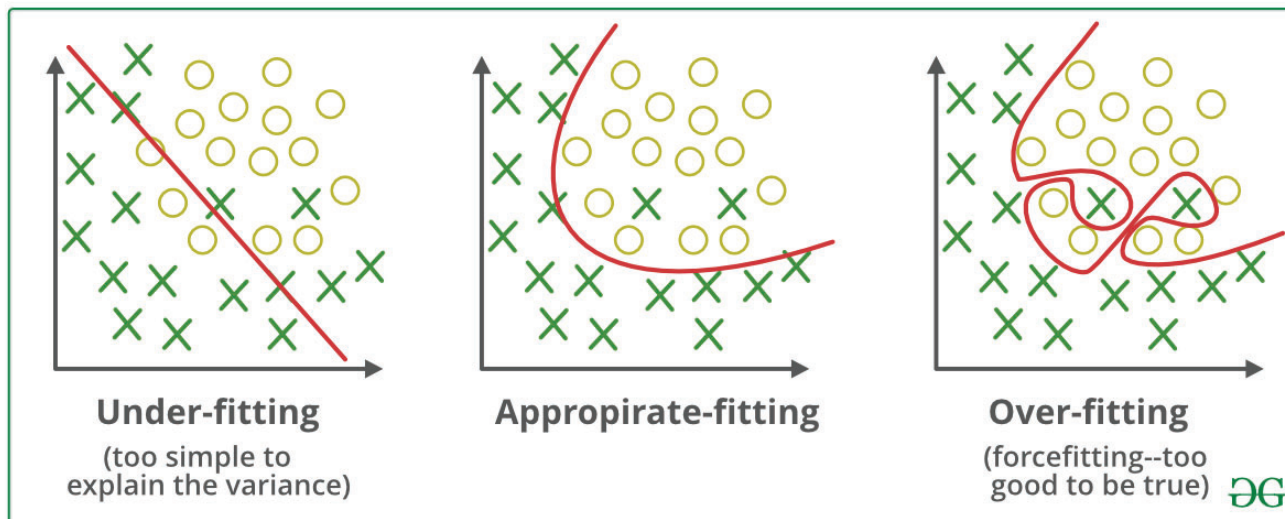
- **WHEN** should a trained model be re-trained?
 - The trained model has performed well on the past examples
 - But at a certain time, the trained model performs significantly poor on the newly coming examples
- **HOW** should a trained model be re-trained?
 - To adapt to the newly coming examples

Generalization capability (1)

- Generalization shows the ability of the model to still achieve high accuracy for future (unseen) data
 - Note: We cannot use any test examples during model selection/training!
 - Use the **validation set** (often extracted from (as a small part of) the original training set) to serve as unseen data in the model training/selection
 - Assumption: The data characteristics are similar between the validation and test sets!

Generalization capability (2)

- 2 common (and should be avoided!) problems of generalization:
 - **Under-fitting:** Achieve low accuracy on all the training, validation and test sets
 - Often make false conclusions (i.e., the “*high bias*” characteristic)
 - **Over-fitting:** Achieve high accuracy on the training set, but low accuracy on the validation and test sets
 - Tend to make different conclusions for the same (or rather similar) examples (i.e., the “*high variance*” characteristic)



(<https://towardsdatascience.com/underfitting-and-overfitting-in-machine-learning-and-how-to-deal-with-it-6fe4a8a49dbf>)

Problem of over-fit learning (1)

- A learned target function h is considered **over-fit** to a specific training set if there exists another target function h' such that:
 - h' produces lower accuracy than h for the training set, but
 - h' produces higher accuracy than h for the whole dataset (including also those examples that are evaluated after the training process)

Problem of over-fit learning (2)

- Assume that D is the whole dataset, and D_{train} the training set
- Assume that $\text{Err}_D(h)$ is the error caused by the target function h on D , and $\text{Err}_{D_{\text{train}}}(h)$ is the error caused by the target function h on D_{train}
- The target function h is over-fit to D_{train} if there exists another target function h' :
 - $\text{Err}_{D_{\text{train}}}(h) < \text{Err}_{D_{\text{train}}}(h')$, and
 - $\text{Err}_D(h) > \text{Err}_D(h')$

Problem of over-fit learning (3)

- The problem of over-fit learning is often caused by:
 - Errors (noises) in the training set (i.e., by a collection/construction of the training set)
 - The number of training examples is too small, or not representative for the overall distribution of all the examples of the learning problem
 - The accuracy is too high/ideal (~100%) for the training set – The training process converges at a target function that is ideal/perfect for the training examples (but not good for future/unseen examples)

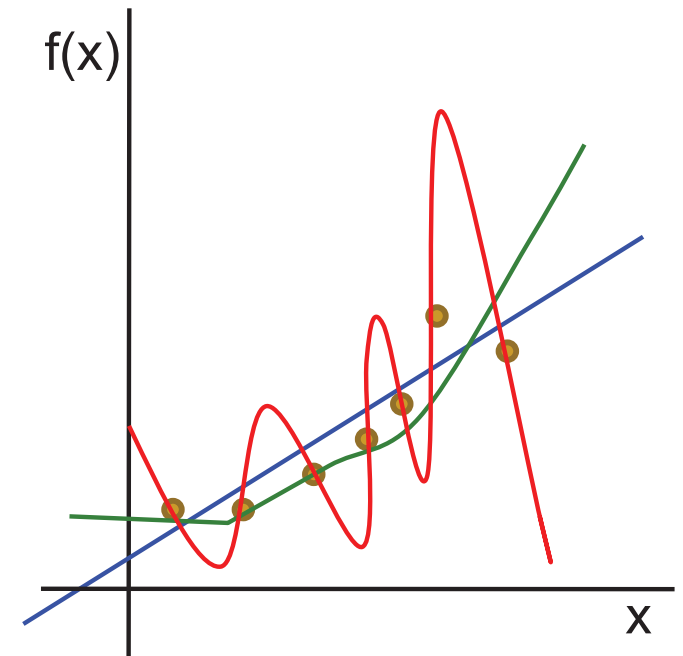
Problem of over-fit learning (4)

- Amongst those target functions learned, which one best generalizes from the training examples?

Important Note: The goal of machine learning is to achieve high accuracy in prediction for future examples, not for the training ones

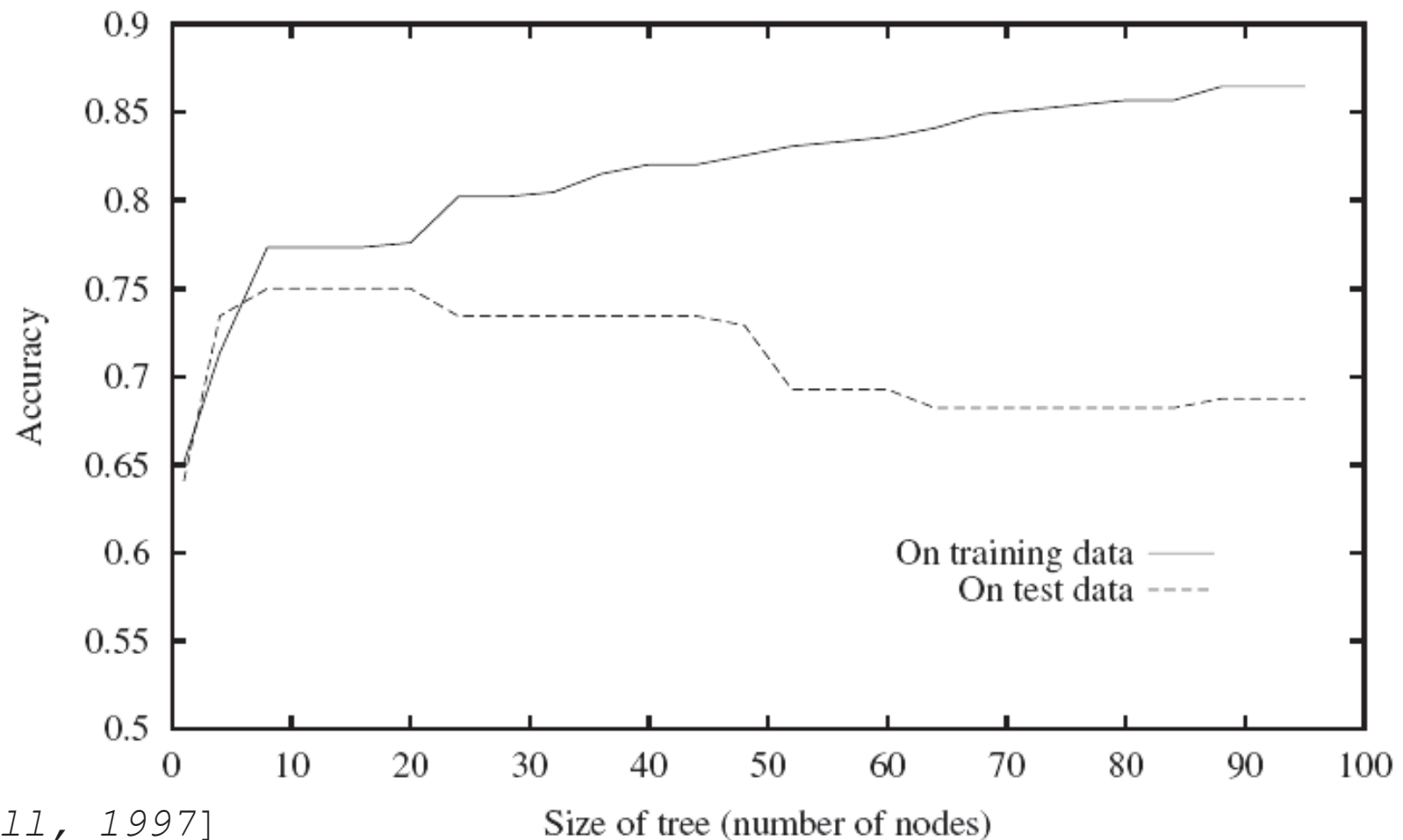
- **Occam's razor:** To select the simplest suitable target function (not necessarily perfect) for the training examples
 - A better generalization
 - Easier for explanation/interpretation
 - Lower in computing cost

Which target function $f(x)$ achieves a highest accuracy for future examples?



Example of over-fit learning

Continuing the Decision Tree learning process decreases the accuracy on the test set though increases the accuracy on the training set



[Mitchell, 1997]

Frameworks and tools for ML (1)

- **TensorFlow** (www.tensorflow.org)
 - ❑ OS: Linux, Mac OS, Windows, Android
 - ❑ Programming language: Python, C++, Java
- **Caffe** (caffe.berkeleyvision.org)
 - ❑ OS: Linux, Mac OS, Windows
 - ❑ Programming language: Python, Matlab
- **Caffe2** (caffe2.ai), **PyTorch** (pytorch.org)
 - ❑ On March, 2018, Caffe2 and PyTorch is merged into a single platform
 - ❑ OS: Linux, Mac OS, Windows, iOS, Android, Raspbian
 - ❑ Programming language: C++, Python
- **Keras** (keras.io)
 - ❑ OS: Linux, Mac OS, Windows
 - ❑ Programming language: Python
- **Theano** (deeplearning.net/software/Theano)
 - ❑ OS: Linux, Mac OS, Windows
 - ❑ Programming language: Python

Frameworks and tools for ML (2)

- **CNTK** (www.microsoft.com/en-us/research/product/cognitive-toolkit/)
 - OS: Windows, Linux
 - Programming language: Python, C++, C#
- **Deeplearning4j** (deeplearning4j.org)
 - OS: Linux, Mac OS, Windows, Android
 - Programming language: Java, Scala, Clojure, Python
- **Apache Mahout** (mahout.apache.org)
 - OS: Any OSs with JVM installed
 - Programming language: Java, Scala
- **MLlib** of Apache Spark (<https://spark.apache.org/mllib/>)
 - OS: Any OSs with JVM installed
 - Programming language: Java, Python, Scala, R
- **Weka** (<http://www.cs.waikato.ac.nz/ml/weka/>)
 - OS: Any OSs with JVM installed
 - Programming language: Java

Online courses

- **Statistics-101** (provided by IBM)
<https://cognitiveclass.ai/courses/statistics-101>
- **Machine Learning with Python** (provided by IBM)
<https://cognitiveclass.ai/courses/machine-learning-with-python>
- **Machine Learning Foundations: A Case Study Approach** (provided by University of Washington)
<https://www.coursera.org/learn/ml-foundations>
- **Machine Learning** (provided by Stanford University)
<https://www.coursera.org/learn/machine-learning>
- **Predictive Analytics and Data Mining** (provided by University of Illinois at Urbana-Champaign)
<https://www.coursera.org/learn/predictive-analytics-data-mining>
- **Data Mining Specialization** (provided by University of Illinois at Urbana-Champaign)
<https://www.coursera.org/specializations/data-mining>

References

- E. Alpaydin. *Introduction to Machine Learning*. The MIT Press, 2004.
- T. M. Mitchell. *Machine Learning*. McGraw-Hill, 1997.
- H. A. Simon. *Why Should Machines Learn?* In R. S. Michalski, J. Carbonell, and T. M. Mitchell (Eds.): *Machine learning: An artificial intelligence approach*, chapter 2, pp. 25-38. Morgan Kaufmann, 1983.